# Dolgov-Kawasaki Stability and Scalaron Mass in Fractional Gravity

David A. Besemer

*Independent Researcher*

david.besemer@sdgft.org

arXiv:SDGFT-2026-15

**Abstract.** We prove the stability of the f(R) = R$^($67/48)$actionundertheDolgov-$
$Kawasakicriterion.Sincetheseconderivativef''(R)isstrictlypositiveforn = 67/48, thetheoryisfreeoft$

## 1 Introduction

Modified gravity theories can suffer from ghost or tachyonic instabilities. We analyze the stability of the SDGFT action.

## 2 Theoretical Formulation

Here we formulate the core mathematical relations governing the system:

$$f''(R) = n(n-1)R^{n-2} > 0 \quad (\text{for } n = \frac{67}{48} > 1) \tag{1}$$

$$m_f^2 = \frac{1}{3}\left(\frac{f'}{f''} - R\right) > 0 \tag{2}$$

## 3 Stability Proof

The positivity of f"(R) guarantees that the scalaron remains a physical particle with a positive mass squared, precluding ghost modes.

## 4 Conclusion

The R$^($67/48)$actionisastableextensionofGeneralRelativity, suitableforbothearly-universeinflationandlate-$
$timecosmology.$

## References

[1] D. A. Besemer, "Axiomatic Foundations of SDGFT," SDGFT-2026-01 (in preparation).

[2] D. A. Besemer, "Effective Spectral Dimension and the Banach Fixed-Point Theorem in SDGFT," SDGFT-2026-04.

[3] D. A. Besemer, `sdgft` Python package, https://github.com/david-besemer/sdgft (2026).